Chebyshev Inequality

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Justification for the Chebyshev Inequality

Consider the definition of the standard deviation,

$$SD = \sqrt{\frac{\sum_{i-1}^{n} (x_i - \bar{x})^2}{n}}.$$

If we square both sides of this equation and multiply both sides of the result by n, we get

$$nSD^2 = \sum_{i-1}^{n} (x_i - \bar{x})^2.$$

This means that the total sum of the squares of the differences from the mean is nSD^2. Now let us consider numbers that are far away from the mean, say x standard deviations or more away. If p is the proportion of numbers in the list that are x or more standard deviations away from the mean, then there are $n * p$ such numbers. The sum of the squares of the differences from the mean for these numbers will be at least

$$np(x * SD)^2 = npx^2SD^2.$$

This cannot be larger than the total sum of the squares of the differences from the mean, $n * SD^2$. Hence

$$npx^2SD^2 \leq nSD^2.$$

Cancelling identical terms and solving for p gives

$$p \leq \frac{1}{x^2}.$$